

Lab 11: Fault-Tolerance

Submit to Gradescope and Autograder by 11:55 pm Thu 4/16

[Starter code](#)
[Qiskit Tutorials](#)

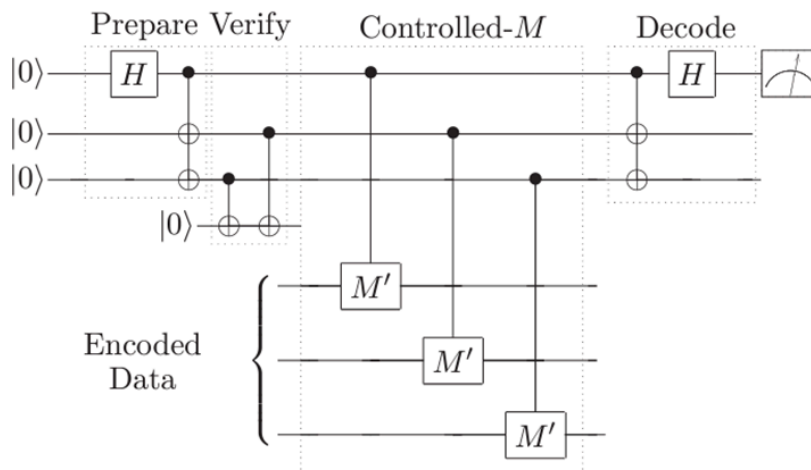
Measurement

We are for the third (and final) time, looking at another way to estimate coordinates on a Bloch sphere. Only this time, we will be considering a logical qubit encoded over 3 physical qubits using the 3-qubit bitflip code, and we will assume that an arbitrary qubit error may occur during our measurement procedure.

Below is the circuits we used in lab 3 to measure the X, Y, and Z axis of a qubit:

Measuring along X-Axis	Measuring along Y-Axis	Measuring along Z-Axis

These circuits are not fault-tolerant, because a single error on q_1 could multiply to several physical qubits in q_0 , rendering error-correction ineffective. The below circuit is a fault-tolerant approach:



1. Prepare the circuit in the special "cat" state $|000\rangle + |111\rangle$ (a multi-qubit generalization of the Bell state).

2. Since an error may occur here and spread to our encoded data later, we must verify this worked by performing 3 parity checks (only one is shown above) between every pair of cat qubits. If any of these measure 1, we should start over at step 1. The [while_loop](#) method will be helpful here
3. Once all the parity checks pass and we are confident that the cat state is prepared, we can apply the appropriate controlled gate transversally across each encoded data bit, ensuring that each cat qubit is only used once to prevent new errors from multiplying.
 - a. X and Z gates should work as expected, as they both have Eigenstates of 1 and -1. Since $1^3=1$ and $(-1)^3=-1$, we don't run into any problems by applying 3 of these controlled gates. But Y's $|0\rangle/|1\rangle$ Eigenvalues are i and $-i$, and $i^3=-i$ and $(-i)^3=i$, so when we apply these gates we'll be off by a phase of -1. Recall from P2 how we can correct this phase. In this case $M'=Y*$ [other sequence of gates to add -1 global phase]
4. The cat qubits are decoded, and the least significant qubit is measured. The more often we measure a 0, the closer we are to a +1 Eigenstate of the axis in question. The more often we measure a 1, the closer we are to a -1 Eigenstate.
5. Since an error could have occurred after the parity checks in step 2, we must repeat this whole process (starting from step 1) 2 more times (for a total of 3). In each iteration, step #4 must write to a different classical register. We can then take a majority vote of the outcomes. Assuming at most one error has occurred, a majority vote should always identify the correct value.

You are to fill in the implementations for `prepare_ancilla`, `FT_meas_[X/Y/Z]`, and `process_counts`. `decode_and_measure` is already completed.

Your code will be run on the same tests as labs 2 and 3, but with additional versions that will insert a random bug somewhere on the ancilla bits. We would need to add a phase-flip code (i.e. make the Shor code) to handle general errors on the data qubits, so we will ignore those now for simplicity. There are also tests to check `prepare_ancilla` and `process_counts`.